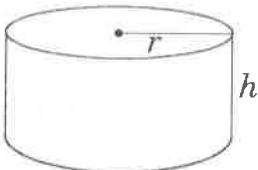


Part 4: Maximize the Volume of a Cylinder

Cylinder 	$A_{\text{base}} = \pi r^2$ $A_{\text{lateral surface}} = 2\pi r h$ $A_{\text{total}} = 2A_{\text{base}} + A_{\text{lateral surface}}$ $= 2\pi r^2 + 2\pi r h$	$V = (A_{\text{base}})(\text{height})$ $V = \pi r^2 h$
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Investigation 1: Maximize Volume of a Cylinder for a Given Surface Area



Your task is to design a cylindrical juice can that uses 470cm² of aluminum. The can should have the greatest volume possible.

- Substitute the given surface area and radius into the surface area formula and solve for the height, h. You can use the rearranged formula ~~SA = 2\pi r^2 + 2\pi r h~~ $h = \frac{SA - 2\pi r^2}{2\pi r}$
- Determine the volume of this can using the formula for the volume of a cylinder: $V = \pi r^2 h$. Record the data in the following table:

Surface Area	Radius	Height	Volume
470	2	35.4	444.8
470	5	10.0	785.4
470	8	1.4	281.5

- What is the maximum volume for the cans in your table? What are the radius and height for this can?

The max. volume is 785.4 cm³.

radius = 5 cm

height = 10 cm

Conclusion: The maximum volume for a given surface area of a cylinder occurs when its height equals its diameter. That is, $h = d$ or $h = 2r$

Example 1:

a) Determine the dimensions of the cylinder with maximum volume that can be made with 600cm² of aluminum. Round the dimensions to the nearest hundredth of a centimeter.

Solution:

We know that a cylinder with height equal to the diameter (double the radius) gives the maximum volume for a given surface area.

To determine the radius of the can with a given surface area, use the surface area formula and substitute $h = 2r$.

Given: $SA = 2\pi r^2 + 2\pi rh$ and $h = \text{diameter} = 2r$

$$SA = 2\pi r^2 + 2\pi r(2r)$$
$$SA = 2\pi r^2 + 4\pi r^2$$
$$SA = 6\pi r^2$$

To solve for the radius given a surface area, you must use the rearranged formula: $r =$

$$\sqrt{\frac{SA}{6\pi}}$$

$$r = \sqrt{\frac{600}{6\pi}} = 5.64$$

The radius of the cylinder should be 5.64 cm and the height should be twice that, or 11.28 cm.

b) Determine the volume of this cylinder

$$\text{Volume} = \pi r^2 h$$
$$\text{Volume} = \pi (5.64)^2 (11.28)$$
$$\text{Volume} = 1127 \text{ cm}^3$$

Apply What You Learned

1. Determine the dimensions of the cylinder with the maximum volume for each surface area. Round the dimensions to the nearest hundredth of a unit.

a) 1200 cm^2

$$r = \sqrt{\frac{SA}{6\pi}}$$

$$r = \sqrt{\frac{1200}{6\pi}}$$

$$r = 7.98 \text{ cm}$$

$$h = d$$

$$= 2r$$

$$= 2(7.98)$$

$$= 15.96 \text{ cm}$$

$$\text{radius} = 7.98 \text{ cm}$$

$$\text{height} = 15.96 \text{ cm}$$

b) 125 cm^2

$$r = \sqrt{\frac{SA}{6\pi}}$$

$$r = \sqrt{\frac{125}{6\pi}}$$

$$r = 2.58 \text{ cm}$$

$$h = d$$

$$= 2r$$

$$= 2(2.58)$$

$$= 5.16 \text{ cm}$$

$$\text{radius} = 2.58 \text{ cm}$$

$$\text{height} = 5.16 \text{ cm}$$

2. Determine the volume of each cylinder in question 1. Round to the nearest whole unit.

$$\begin{aligned} \text{a) } V &= \pi r^2 h \\ &= \pi (7.98)^2 (15.96) \\ &= 3192.92 \text{ cm}^3 \end{aligned}$$

$$3193 \text{ cm}^3$$

$$\begin{aligned} \text{b) } V &= \pi r^2 h \\ &= \pi (2.58)^2 (5.16) \\ &= 107.9 \text{ cm}^3 \end{aligned}$$

$$108 \text{ cm}^3$$